PERFECT COMPLEXES ARE RELATIVELY PERFECT, AND DERIVED TENSORING PRESERVES RELATIVE PERFECTIVITY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12383-12391 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12393-12399, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Definitional-reduction template; sibling of L16175.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma affine-locally-rel-perfect; Derived Categories of Schemes, lemma perfect-relatively-perfect.

Source reference. Original lemma 0DKQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12313–12335. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DI4: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12399–12399. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S which is flat and locally of finite presentation. A perfect object of $D(\mathcal{O}_X)$ is Y-perfect. If $K, M \in D(\mathcal{O}_X)$, then $K \otimes^{\mathbf{L}}_{\mathcal{O}_X} M$ is Y-perfect if K is perfect and M is Y-perfect.*

Proof. Reduce to the case of schemes using Lemma 0DKQ and then apply Derived Categories of Schemes, Lemma 0DI4. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.